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Politecnico di Milano

03rd September 2026

Milan, Italy

Philosophy of Science and Technologies examination - Paper 3

In this document, I discuss the topic of additional tools of science presented by Antoine Brandelet during the course of Philosophy of Science and Technologies, held at Politecnico di Milano in the academic year 2025/26. In particular, I address Computer Simulations.

The assignment pursued is the following : *Explain why scientific practice goes (and sometimes needs to go) beyond the traditional trichotomy of theories, observations, and experiments in such a way to employ different tools: from tools of thought like thought experiments all the way to technological tools like computer simulations. Choose one of these tools and present the main philosophical issues at stake. Describe two philosophical approaches that promise to tackle such issues. Defend your favorite one and criticize the other, by referring to concrete case studies taken from any scientific field.*

Traditional trichotomy of science and its limitations

In the first chapter of the course we discussed the problem of demarcation of science. The current of thoughts were different, some more experimentalist and other more rationalist, however a “traditional trichotomy” can be highlighted with foundation of theories, gathering of observations, experimental results, and deductions or inferences to check/formulate theories.

Experiments are fundamental. They come from a need of isolating a process (under investigation) from effects of other processes; and they permitted to infer many phenomena which were otherwise invisible to our senses; thus, expanding scientific knowledge and formulating, confirming or falsifying laws. For the sake of this chapter, we think about experiments as “intervene and observe”: scientists prepare the setup for the experiment, they execute the experiment, observe the results, they intervene to change the setup and observe what changes in the outcome.

Example: Performing Young’s double slit experiment, which was fundamental to develop Quantum Mechanics, one runs the experiment at first with light rays (photons), then tries to see what happens with a beam of electrons, then sends one electron at each time, then modifies the distances between the two holes, etc..

Isolating a phenomenon is physically happening. To experiment on phenomenon A one needs to be able to replicate phenomenon A under many circumstances, repeatedly and manipulate it. Unfortunately, this happens to be unfeasible in many situations.

Example: a scientist who aims at studying populations evolution should isolate the population of animals (or people), take different samples, operate many populations, and then see what happens when he changes the system. This can be unpractical, unfeasible and unethical.

Somehow, I would judge this as a consequence of the boundedness of our existence. We are limited in our capabilities, in our resistance to extreme conditions and in time. That is, humans cannot explore in first person the whole regions of reality to collect knowledge. Therefore, scientists have developed other tools such as Thought Experiments, Models, and Computer Simulations.

Sometimes experiments may also be feasible, but using different tools ends up being easier, cheaper and faster. In science there is also a need for immediate results due to economical reasons, therefore it is important to study the validity of such methods which otherwise could be misused irrationally, even by scientists.

Models & thought experiments – briefly

Before talking about Computer simulations, I think it is fundamental to give a quick overview of my position on models and thought experiments TE.

Thought experiments are experiments happening in “the laboratory of the mind”, and without any physical implementation the result is coming from the theoretical assumptions of the theory. These are often used to challenge some fundamental concepts (example is the Schoedinger’s cat).

Models are instead a more rigorous mathematical description of the behaviour of properties of some phenomena. From a phenomenon X, one writes its model in mathematical language to predict the evolution of its main properties observed. Models represent mono-directionally the reality, in a more simple and tractable way.

Computer simulations

Computer simulations (CS) are used to get quantifications, predict behaviours and test hypotheses. A code is written in some programming language, then it is executed by an electronic circuit and the result can be a scalar quantity, a configuration, or a visual representation, etc.

The goal of CS is to “simulate” reality, reproduce it so that we can “experiment” with the simulation. Is the use of CS legit and rigorous? What is the relation between such results and reality? How is a sequence of calculations executed on a circuit, representing an ontological phenomenon? If new knowledge comes from a CS, how can one justify its validity and relevance for scientific laws?

Similarities between computer simulations and experiments

CS may be interpreted as experiments E because there is an analogy. After writing the model of the phenomena, a code is written, executed and the results are interpreted; then one changes slightly the initial conditions and re-runs the software to observe changes, and so on. In this sense, simulations reproduce the experimental structure of “intervene and observe”.

Moreover, the connection between experiments and CS can be interpreted as epistemological because the experimenter and the simulationer will have the same experience interacting with a computer and with an experimental apparatus :

“The results of a simulation are unpredictable. [...] The complexity and unpredictability of a computer program invite the scientist to present it as a black box, interacting with it in an experimental manner, trying things out to see what will happen” (Dowling, 1999)

In support of this view, both CS and experiments operated not on the target T, describe T through a model. The model promises to give as output the same behaviour of analogous properties in T and in the experiment/CS.

The fundamental question is, if E can produce new knowledge, can CS do the same?

The work of Julie Jebeile supports a positive answer with a more complicated view. First, it is noticed that both experiments and CS provide areas of exploration, and there is an intervention, justification of inferences, development over time and can function as black boxes. If the model is in common between CS and E, they can both bring novelty according to the *first-time criterion*. That is, they can provide new knowledge only if it is the first time that such piece of knowledge is added in a “relevant way” to the existing knowledge. Moreover, it is specified that this doesn’t involve the so-called *terra-incognita context*, that is situations in which data is scarce, there is a lack of theories and thus model on which to write the simulation; the simulation cannot be executed.

Differences between computer simulations and experiments

The different nature is argued instead by Claus Beisbart.

“Interventions” on the simulations are considered not to be real interventions in the same sense as experiments, plus there is no intervention on the hardware and either on the target. The conclusion is that the structure “intervene-observe” as not the same nature as in experiments. Moreover, CS are *over-controlled* by model assumptions, which means that the experiment can “execute the model” and provide calculations deriving from the model itself, not bringing any new information.

Example: recall young’s double slit experiment. If the phenomena is described by Classical Mechanics, where electrons are modelled as particles, then no CS simulation on such model will ever observe the wave nature.

Thus, the epistemic power is considered to be different and CS can work only as surrogates for experiments. CS are a representation of the experiments, which can in their turn produce knowledge, but they do not coincide epistemologically.

Example: in drugs studies, simulations can be executed to find new medicines, but one will only employ this because cheaper and faster. Then, in-vitro experiments will be necessary to test how to produce the desired drug. The CS only worked to discard the range of assumptions that will most probably not work, and efficient the process.

Personal considerations

I think that CS do not have the same epistemological power as experiments.

However, I think that experiments on the target T differ from experiments on a representative R, where I denote as “representative” the system R on which the experiment is executed assuming that the model M describes the same causal relationships in R that are there in T, for some target-properties.

If experiments outcome on T differ from prospected results by M, then either M is wrong (in assumptions or causal relationships), or the theory behind the model is wrong. However, if experiments outcome on R differ from the prospected results by M, it may be that R is not correctly represented by M, thus the situation becomes more complicated. Indeed, results on R may not coincide with results on T, this can be because of a wrong assignment M-T or M-R.

Example: In Formula 1 competitions it is known that results from the wind galleries may not coincide with results on the road, the knowledge from the gallery is not correct for the real target that is the phenomenon during the race.

Thus, already inside the set of all experiments we have different epistemological values.

CS are a series of computations run on an electronic circuit, which is in my opinion, only an execution of the model. They only operate a deductive reasoning that starts from modelling assumptions, which is not generating any new knowledge. Monte-Carlo simulation, make use of stochastic processes and the computation seem to be not deductive, but this can also be formally developed by a probabilistic theory where the objects described are the whole distributions on a large number of samples. Because of this I claim that CS are actually an extension of brain capacity to develop TE.

Example in favour of the claim: imagine that Galileo had the possibility to access a CS to test different situations for the law of falling bodies. If he had tested the different scenarios of two bodies A,B with masses $m_A > m_B$, and then he would have tested the body A+B with mass m_A+m_B , he would have obtained the same results as he did with his TE.

CS seem to place themselves in a very peculiar situation. They are an execution of the model, which is what is operated already by TE, however they permit an experimentalist interaction with it. Because of this, they only use deductive reasoning on the model they describe.

Let me frame my thoughts in a more linear way now. Experiments on the target differ in their epistemological value from experiments in a representative. Some of the representative's properties are described by the same model as the main target properties. However, these properties will be rarely completely independent from the rest of the target, and this model will be only an approximation. Therefore, experiments on a representative represent simply more executions of the model, which does not differ from what CSs are doing: a series of deductive reasonings (execution) on the model. Where, because the reasoning is deductive and truth preserving (AND, OR, and NOT operations) the knowledge is not expanded and already contained in the premises.

Example: Imagine you want to study ocean circulation and its role in Earth's climate using as representative a rotating tank of water. This tank of water will be heated in some parts and cooled down in others to simulates what happens in a real scenario. The experiment will be able to reproduce many mechanisms such as convection, differences in the density of water, etc.; thus one could observe the currents formed. However, the tank is trying to simulate the real ocean as one would do with a CS:

assuming finite amount of water confined, then you make it rotate and one part is heating up while the other cooled down. A part from numerical errors, the two tools will describe the same reality, implementing the same model and neither of them will have a full epistemological power on the ocean's laws.

References:

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